

F78PB Probability and Statistics B

Hypothesis Testing

Lead pollution example continued

Example

Find the equation of the least squares regression line for the example on lead pollution on vegetation.

Residuals

The difference between the response y_i and the fitted value $\hat{y}_i$

$$e_i = y_i - \hat{y}_i = y_i - \hat{\alpha} - \hat{\beta}x_i, \quad i = 1, 2, \ldots, n$$

is called the **residual at x_i.** The residuals estimate the true errors

$$\varepsilon_i = y_i - \alpha - \beta x_i, \quad i = 1, 2, \ldots., n$$

Properties

$$\sum_{i=1}^{n} e_i = 0$$

$$\sum_{i=1}^{n} e_i x_i = 0$$

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} y_i^2 - \hat{\alpha}\sum_{i=1}^{n} y_i - \hat{\beta}\sum_{i=1}^{n} x_i y_i = s_{yy} - \frac{s_{xy}^2}{s_{xx}}$$

(Use this form in $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2$)

Lead pollution example continued

Example

Find the value of $\hat{\sigma}^2$ for the example on lead pollution on vegetation.

We first calculate

$$s_{yy} - \frac{s_{xy}^2}{s_{xx}} = 2653.75 - \frac{3527.5^2}{5561} = 416.157.$$

Then:

$$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2 = \frac{1}{n-2}(s_{yy} - \frac{s_{xy}^2}{s_{xx}}) = \frac{1}{14} \times 416.157 = 29.726$$

Variability Explained by the Model

For each data point y_i:

$$y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}).$$

Interpretation:

- Decompose deviation from the mean into:
  - Residual $(y_i - \hat{y}_i)$: unexplained by the model.
  - Explained part $(\hat{y}_i - \bar{y})$: due to regression.

This is the basis for partitioning variability.

Squared Decomposition

Squaring and summing:

$$\sum_i (y_i - \bar{y})^2 = \sum_i (y_i - \hat{y}_i)^2 + \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}).$$

The cross-term vanishes because:

$$\sum_i (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) = \hat{\beta} \sum_i e_i (x_i - \bar{x}) = 0.$$

Result:

$$\sum_i (y_i - \bar{y})^2 = \sum_i (y_i - \hat{y}_i)^2 + \sum_i (\hat{y}_i - \bar{y})^2.$$

Sum of Squares Definitions

$$SS_T = \sum_i (y_i - \bar{y})^2 \quad \text{Total variability}$$

$$SS_{RES} = \sum_i (y_i - \hat{y}_i)^2 \quad \text{Residual (unexplained)}$$

$$SS_{REG} = \sum_i (\hat{y}_i - \bar{y})^2 \quad \text{Regression (explained)}$$

Interpretation:

- SS_T: overall spread of y.
- SS_{REG}: variability captured by the model.
- SS_{RES}: leftover variability.

Visual Illustration

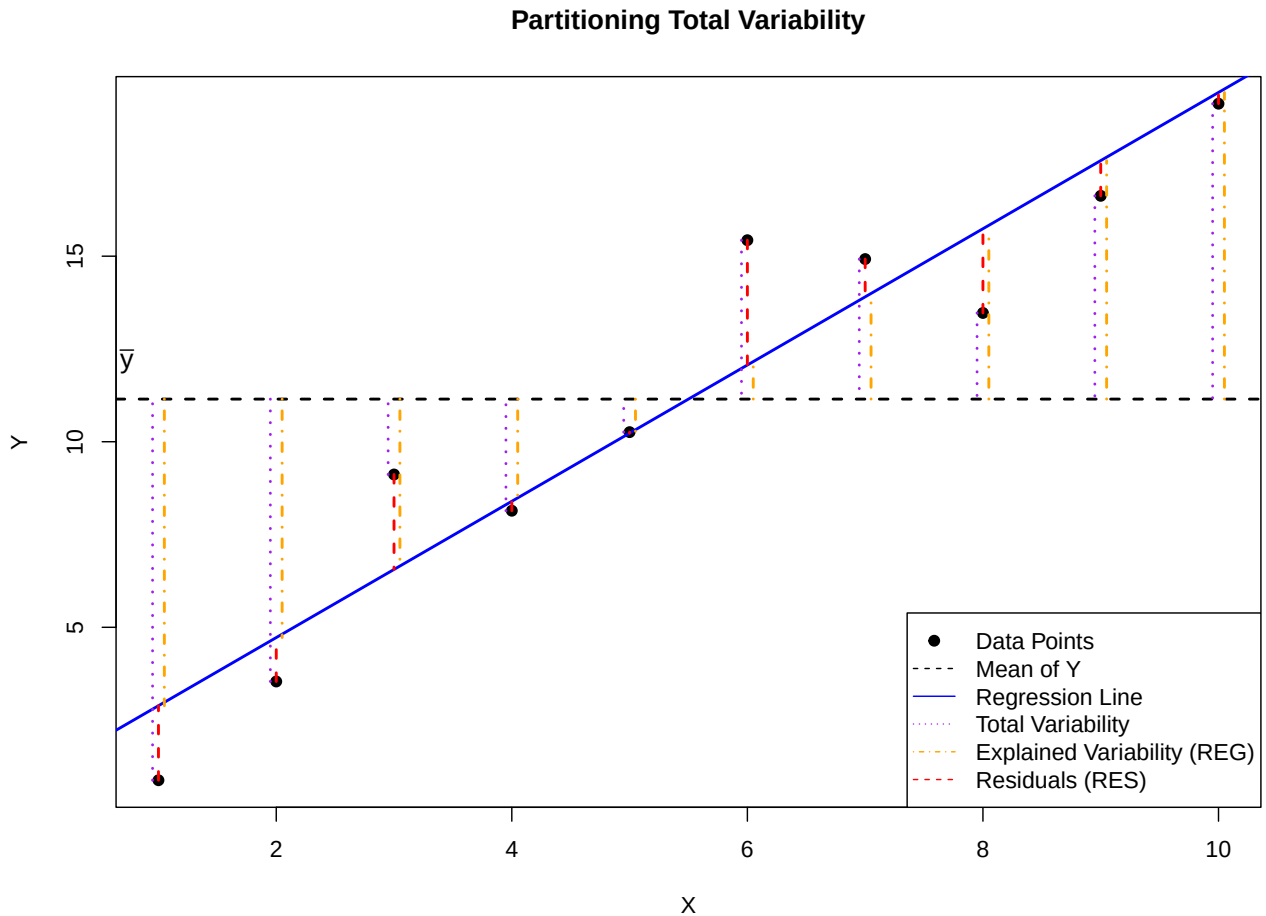


Figure: Partitioning of variability: Total (purple), Explained (orange), Residual (red).

Calculation Details

$$\begin{aligned} \mathrm{SS}_{\mathrm{T}} &= s_{yy}, \\ \mathrm{SS}_{\mathrm{REG}} &= \hat{\beta}^2 s_{xx}, \\ \mathrm{SS}_{\mathrm{RES}} &= s_{yy} - \frac{s_{xy}^2}{s_{xx}}. \end{aligned}$$

Remark: Alternatively, compute $\mathrm{SS}_{\mathrm{RES}}$ directly:

- Find fitted values $\hat{y}_i$.
- Compute residuals $y_i - \hat{y}_i$.
- Square and sum.

Coefficient of Determination

$$R^2 = \frac{\mathrm{SS}_{\mathrm{REG}}}{\mathrm{SS}_{\mathrm{T}}} = \frac{s_{xy}^2}{s_{xx}s_{yy}}.$$

Interpretation:

- Proportion of variability explained by the model.
- For simple linear regression: $R^2 = r^2$ (sample correlation squared).

Full Normal Model

Assume:

- Errors ε_i are independent, and $\varepsilon_i \sim \mathrm{N}\left(0, \sigma^2\right)$.
- Then $Y_i \sim \mathrm{N}\left(\alpha + \beta x_i, \sigma^2\right)$.

Implications:

- Can write likelihood for $\underline{y}$.
- MLE for α, β same as least squares.
- MLE for σ^2: $\frac{n-2}{n}\widehat{\sigma^2}$.

Inference on β

Under full normal model:

$$\frac{(n-2)\widehat{\sigma}^2(\underline{x},\underline{Y})}{\sigma^2} \sim \chi^2_{n-2}, \qquad \frac{\hat{\beta}(\underline{x},\underline{Y}) - \beta}{\sqrt{\widehat{\sigma}^2(\underline{x},\underline{Y})/s_{xx}}} \sim t_{n-2}.$$

Estimated standard error of $\hat{B}$:

$$\sqrt{\widehat{\sigma}^2(\underline{x},\underline{Y})/s_{xx}}.$$

Uses:

- Confidence intervals for β.
- Hypothesis tests on β.

Lead pollution example continued

Example

Find the 95% confidence interval for β in the example on lead pollution on vegetation.
Also test the null hypothesis that the slope coefficient is equal to 0 against the alternative that it is not equal to 0.

Properties of $\hat{\alpha}$ and inference about α

Remember: $\hat{\alpha} = \bar{y} - \hat{B}\bar{x}$ (Least Squares and Maximum Likelihood Estimator).

We can prove that:

1. $\mathbb{E}(\hat{\alpha}(\underline{x}, \underline{Y})) = \alpha$ (unbiased estimator)
2. $\mathbb{V}\text{ar}(\hat{\alpha}(\underline{x}, \underline{Y})) = \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}})$

Under full normal model

$$\hat{\alpha}(\underline{x}, \underline{Y}) \sim \text{N}\left(\alpha, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

and

$$\frac{\hat{\alpha}(\underline{x}, \underline{Y}) - \alpha}{\hat{\sigma}(\underline{x}, \underline{Y})\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}}} \sim t_{n-2}$$

which can be used for hypothesis tests and CIs.

Assumptions and Validity

- Recall we assumed $x_1, \ldots, x_n$ are constant and known.
- In practice this is often not the case (as in the lead pollution example)
- We still use the same methodology

For validity we focus on the assumptions: i.e. the ε_i values are independent and $\varepsilon_i \sim \mathrm{N}\left(0, \sigma^2\right)$. There are three features here:

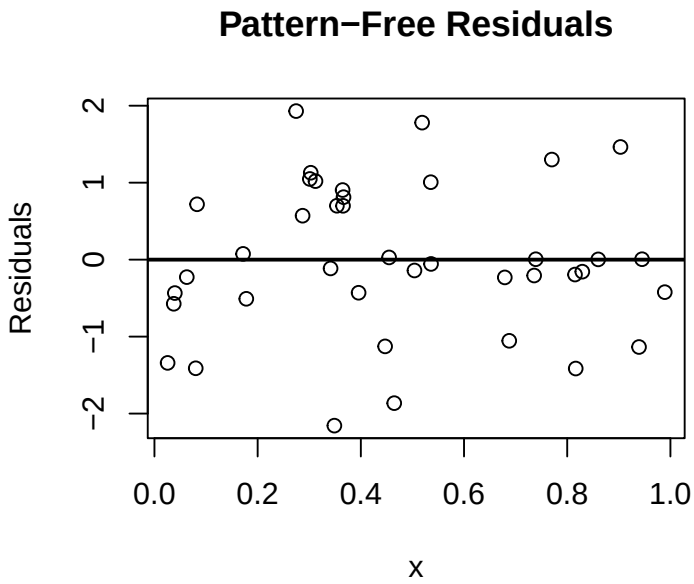
- independence
- normality
- constant variance

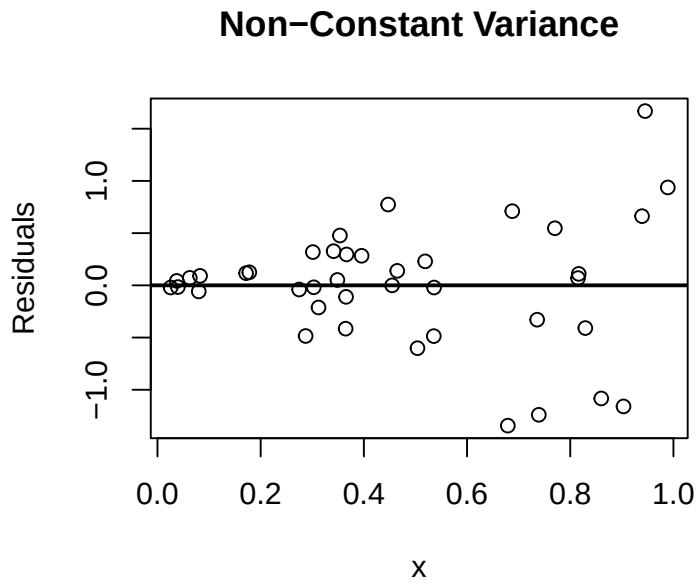
To check these we use 'residual plots':

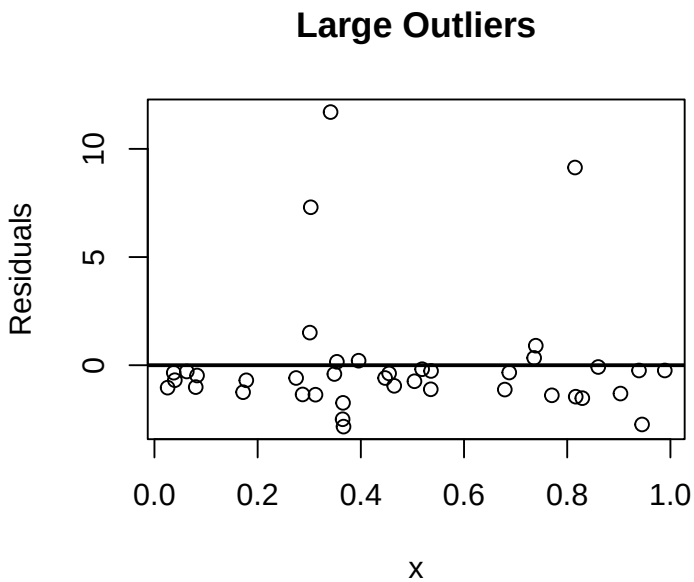
- for normality use a dotplot / histogram;
- for independence and constant variance use a 'residuals v. fitted values' plot.

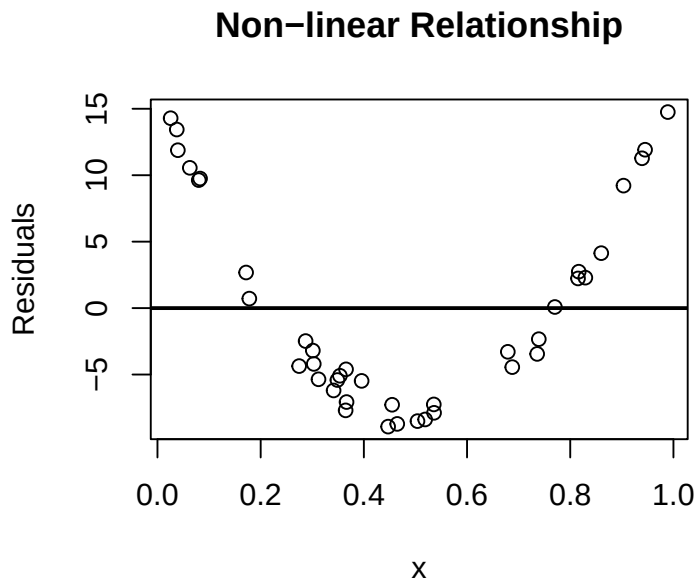
Here we want a random or 'pattern-free' plot.

Examples


Pattern−Free Residuals
Residuals
x
−2
−1
0
1
2
0.0
0.2
0.4
0.6
0.8
1.0


Non−Constant Variance
Residuals
x
−1.0
0.0
1.0
0.0
0.2
0.4
0.6
0.8
1.0


Large Outliers
Residuals
x
0
5
10
0.0
0.2
0.4
0.6
0.8
1.0


Non−linear Relationship
Residuals
x
−5
0
5
10
15
0.0
0.2
0.4
0.6
0.8
1.0

Prediction

- Suppose we have a new value x_0 then our prediction is

$$\hat{y}(x_0) = \hat{\alpha} + \hat{\beta}x_0$$

- What is the accuracy of this prediction?
- It depends on what we are predicting:
  1. The expected (or mean) Y at x_0, i.e.

$$\mathbb{E}[Y(x_0)] = \alpha + \beta x_0$$

  2. An individual value of Y at x_0, i.e.

$$Y(x_0) = \alpha + \beta x_0 + \varepsilon_0$$

- Both have the same estimate but the later is subject to more variation.
- The standard error of the estimate is:

For the expected Y at x_0:

$$\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

For an individual value of Y at x_0

$$\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} + 1}$$

Confidence intervals for prediction

1. For the expected Y at x_0 the limits of a $100(1-\alpha)\%$ confidence interval are:

$$\hat{\alpha}(\underline{x},\underline{Y}) + \hat{\beta}(\underline{x},\underline{Y})x_0 \pm t_{n-2,\frac{\alpha}{2}}\hat{\sigma}(\underline{x},\underline{Y})\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{s_{xx}}}$$

2. For an individual value of Y at x_0 the limits of a $100(1-\alpha)\%$ confidence interval are:

$$\hat{\alpha}(\underline{x},\underline{Y}) + \hat{\beta}(\underline{x},\underline{Y})x_0 \pm t_{n-2,\frac{\alpha}{2}}\hat{\sigma}(\underline{x},\underline{Y})\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{s_{xx}}+1}$$